

SATHISH PALANISAMY

Junior Full Stack Developer

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PROFESSIONAL SUMMARY

Full Stack Developer with 1+ year of experience building scalable web and mobile applications using JavaScript, PHP, and Python. Skilled in React.js, TypeScript, Node.js, FastAPI, and REST API development. Experienced with microservices architecture, JWT authentication, MVC frameworks such as Laravel and CodeIgniter, and building Retrieval-Augmented Generation (RAG) systems. Strong expertise in MySQL and MongoDB, with a focus on performance optimization and clean code practices.

CORE COMPETENCIES

Full Stack Web Development • RESTful API Design & Integration • Mobile App Development (React Native) • Microservices Architecture • Real-Time Systems (Socket.io) • Authentication & Authorization (JWT) • Retrieval-Augmented Generation (RAG) Systems • FastAPI • Relational & NoSQL Database Design • Performance Optimization • Clean Code & MVC Design Patterns

TECHNICAL SKILLS

- **Programming Languages:** JavaScript, Python, PHP, Java, SQL
- **Frontend:** React.js, TypeScript, Angular, Tailwind CSS, HTML5, CSS3
- **Backend:** Node.js, Express.js, RESTful API, PHP, Python, FastAPI
- **Frameworks:** Laravel, CodeIgniter, React Native
- **Databases:** MySQL, MongoDB
- **Tools & Technologies:** Git, Postman, TanStack Query, Socket.io, JWT, VS Code, Cursor AI

PROFESSIONAL EXPERIENCE

Junior Full Stack Developer | Sparkle Digital Solutions Pvt. Ltd.

Oct 2025 – Present

Designed and built a School ERP system covering payroll, HR, attendance, and leave management for 200+ users. Owned the project end-to-end — back-end development, mobile app development, API integration, real-time communication, and database design — in a live production environment.

Backend Development (CodeIgniter & Laravel)

- Architected scalable server-side modules using **CodeIgniter and Laravel** (MVC) for payroll, attendance, leave management, and role-based access control.
- Built reusable service layers with robust error handling and input validation, following clean code practices to ensure a secure, production-ready system.

Mobile Application Development (React Native)

- Built a cross-platform **React Native** mobile app for staff and management, covering payroll, attendance, and HR features with a responsive interface.
- Integrated RESTful APIs for real-time sync and implemented **JWT authentication** with persistent session handling across Android and iOS.

RESTful API Development & Integration

- Created secure RESTful APIs with JWT authentication, middleware validation, rate limiting, and structured error handling across web, mobile, and back-end services.
- Integrated 3+ third-party services for real-time notifications and automated reporting, with clear API documentation that improved cross-team efficiency by 30%.

Node.js Microservices & Real-Time Communication (Socket.io)

- Delivered modular Node.js microservices for background tasks and inter-service communication, keeping the system decoupled and scalable.
- Used Socket.io for live notifications, real-time attendance updates, and dynamic dashboard refresh.

API Testing, Caching & Database Management

- Tested APIs with Postman and implemented **TanStack Query** for smart caching, reducing API calls by 35% and improving app responsiveness.
- Managed MySQL operations, including schema design, query optimization, indexing, and migrations, for a reliable, high-performing database.

AI-Powered Data Retrieval (RAG System)

- Built and deployed a **Retrieval-Augmented Generation (RAG)** system within the School ERP, served through a **FastAPI** backend, to enable natural-language data retrieval.
- Enabled staff to retrieve student data and school owners to retrieve day-to-day income and employment details through natural-language queries.
- Integrated the RAG service with the existing ERP data layer so results reflect live, up-to-date records.

Full Stack Developer Intern | Let's Grow More

Apr 2024 – Jun 2024

- Built fully responsive web pages with HTML, CSS, JavaScript, and React.js, focusing on clean UI, cross-browser compatibility, and a smooth user experience across all screen sizes.
- Implemented server-side functionality using Node.js and Express.js, developing RESTful API endpoints to handle data flow between front-end and back-end services.
- Implemented dynamic rendering, form validation, and real-time feedback to boost user engagement and overall application functionality.
- Collaborated with the team through code reviews and Git-based version control, gaining hands-on experience with real-world development workflows and best practices.

PROJECTS

Quick Cart – E-commerce Grocery Web Application

Personal full-stack project: a grocery e-commerce platform covering the core shopping flow from browsing to checkout, built end-to-end on the MERN stack.

- Built the application using the **MERN stack** (MongoDB, Express.js, React.js, Node.js) with a component-based front end and a RESTful Express/Node.js back end.
- Implemented user authentication and session handling to support secure account access.
- Developed a product catalog module for browsing and viewing grocery items.
- Built cart management functionality, including adding, updating, and removing items.
- Implemented the order placement flow to let users complete purchases from cart to checkout.
- Designed a responsive UI so the app works smoothly across different screen sizes.
- Optimized real-time database read/write operations in MongoDB to keep the app performant and scalable as data grew.

Tech: MongoDB, Express.js, React.js, Node.js

Chat Application (MERN Stack + Socket.IO)

Personal full-stack project: a real-time messaging application focused on instant, secure user-to-user communication.

- Built a real-time chat platform using the **MERN stack** with **Socket.IO** for instant, bidirectional user-to-user messaging.
- Implemented JWT-based authentication to secure user login and protect chat sessions.
- Used MongoDB to persist user accounts and message history, enabling chat history to load across sessions.
- Structured the front end in React.js for a responsive, real-time chat interface.

Tech: MERN Stack, Socket.IO, JWT

EDUCATION

Bachelor of Technology (B.Tech) — Information Technology

2021 – 2025

Government College of Technology, Coimbatore

CGPA: 7.25

ADDITIONAL INFORMATION

- **Languages:** English, Tamil
- **Certifications:** Full Stack Development – Novi Tech (2024); React.js Development – Novi Tech; AI Facial Detection – Novi Tech
- **Activities:** Exterro Java Workshop – learned Java, OOP, and algorithms; Clean Coding Workshop – focused on SOLID principles; Web Development Using React.js – built apps with React.js